

Efficiency–TFP Analysis

Portugal (Santos et al., 2021)

MAPS Project — Deliverable 4.2 (WP4: Improving Integrated Assessment Modelling)

Period: 1960–2014 • Observations: 55 • Date: 22 May 2026

This report documents the Efficiency–TFP cointegration analysis for Portugal (Santos et al., 2021) (1960–2014). Following De Ketelaere et al. (2026) and Santos et al. (2021), it tests whether changes in Total Factor Productivity (TFP) are explained by improvements in final-to-useful exergy efficiency, and assesses the stability of useful exergy intensity for scenario analysis.

Steps of the Analysis

1. Output elasticities (a_K , a_L) from income data
2. Labour inputs: unadjusted and quality-adjusted
3. TFP: unadjusted and quality-adjusted (z-normalised to base year)
4. Exergy efficiency and useful exergy intensity
5. Unit root tests (ADF and PP) on TFP and efficiency series
6. VAR estimation and residual diagnostics
7. Johansen cointegration tests: H_z (no deterministic) and H_c (restricted constant; no trend)
8. Useful exergy intensity constancy assessment

Key Results

i) Long-run TFP–Efficiency relationship [H_c specification — restricted constant; no trend]:

$$TFP = EFF^{1.0480} \quad (\text{quality-adjusted TFP})$$

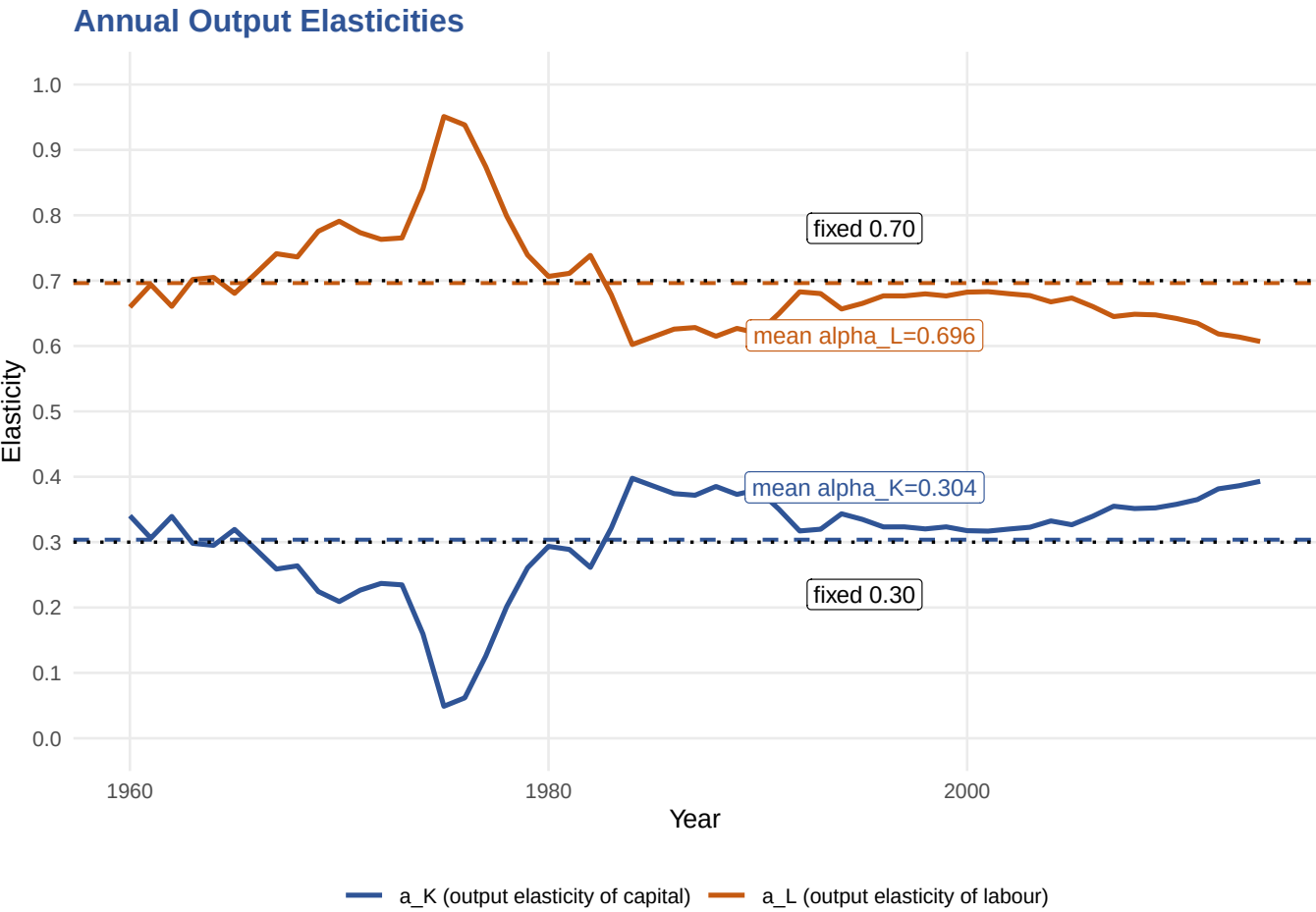
(z-form: $z_{TFP} = 1.0480 \times z_{EFF}$)

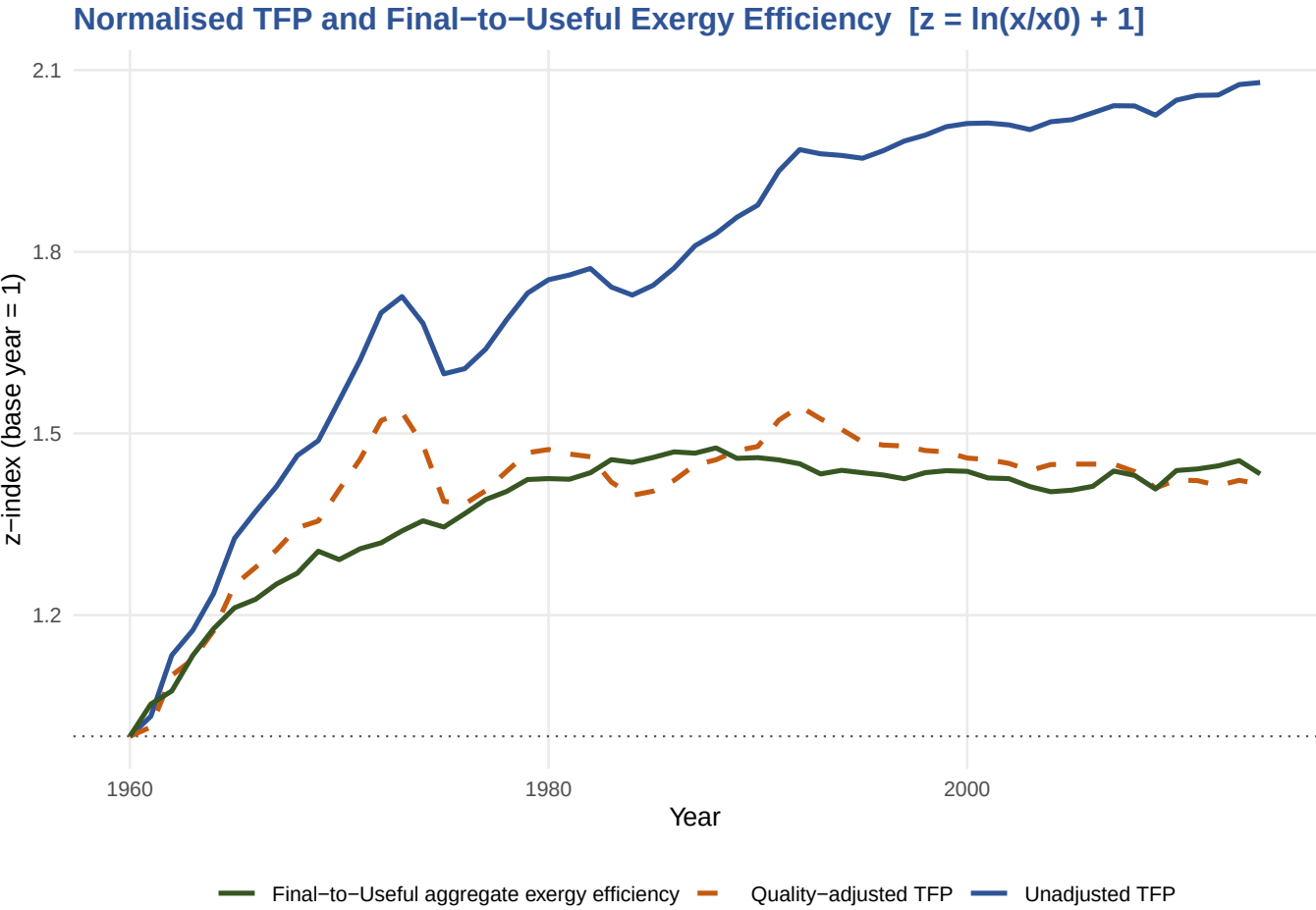
ii) Aggregate production function (Cobb–Douglas):

$$Y = K_{services}^{0.3038} \times hL^{0.6962} \times EFF^{1.0480}$$

iii) Useful exergy intensity:

Not constant. Trend: 0.428% p.a. ($R^2=0.313$)





Unit Root Tests — ADF and PP (5% significance level)

Series: z_tfp_unadj | Verdict: I(1) | ADF: I(1) | PP: I(1)

Test	Specification	Statistic	CV 1%	CV 5%	CV 10%	Aux.	Val.	H0 rej.?	Verdict
ADF	Levels (trend+intercept)	−3.3419	−4.1305	−3.4921	−3.1748	Lags (BIC)	3	No	I(1)
ADF	1st diff. (intercept)	−5.6182	−3.5482	−2.9126	−2.5940	Lags (BIC)	0	Yes	—
PP	Levels (trend+intercept)	−3.2836	−4.1213	−3.4878	−3.1723	BW (NW)	3	No	I(1)
PP	1st diff. (intercept)	−5.6644	−3.5482	−2.9126	−2.5940	BW (NW)	4	Yes	—

Series: z_tfp_adj | Verdict: I(1) | ADF: I(1) | PP: I(1)

Test	Specification	Statistic	CV 1%	CV 5%	CV 10%	Aux.	Val.	H0 rej.?	Verdict
ADF	Levels (trend+intercept)	−2.8730	−4.1525	−3.5024	−3.1807	Lags (BIC)	1	No	I(1)
ADF	1st diff. (intercept)	−3.4811	−3.5527	−2.9145	−2.5950	Lags (BIC)	2	Yes	—
PP	Levels (trend+intercept)	−3.3874	−4.1213	−3.4878	−3.1723	BW (NW)	4	No	I(1)
PP	1st diff. (intercept)	−5.4994	−3.5482	−2.9126	−2.5940	BW (NW)	4	Yes	—

Series: z_eff | Verdict: I(1) | ADF: I(1) | PP: I(1)

Test	Specification	Statistic	CV 1%	CV 5%	CV 10%	Aux.	Val.	H0 rej.?	Verdict
ADF	Levels (trend+intercept)	−2.8459	−4.1243	−3.4892	−3.1731	Lags (BIC)	1	No	I(1)
ADF	1st diff. (intercept)	−5.4869	−3.5482	−2.9126	−2.5940	Lags (BIC)	0	Yes	—
PP	Levels (trend+intercept)	−2.5647	−4.1213	−3.4878	−3.1723	BW (NW)	4	No	I(1)
PP	1st diff. (intercept)	−5.7737	−3.5482	−2.9126	−2.5940	BW (NW)	4	Yes	—

[OK] All series are I(1): unit roots confirmed in levels, stationary in first differences.

Green light — Johansen cointegration analysis is appropriate.

VAR Residual Portmanteau Tests

VAR(2): TFP_unadj ~ EFF | T=58 (1962–2019)

Lags	Q-Stat	Prob.	Adj Q-Stat	Prob.	df
h=1	0.110059	---	0.111990	---	---
h=2	6.610195	---	6.844273	---	---
h=3	13.022210	0.0112	13.606040	0.0087	4
h=4	16.293600	0.0384	17.119750	0.0289	8
h=5	16.864070	0.1548	17.744040	0.1237	12

* Test is valid only for lags larger than the VAR lag order.

VAR(2): TFP_adj ~ EFF | T=58 (1962–2019)

Lags	Q-Stat	Prob.	Adj Q-Stat	Prob.	df
h=1	0.080542	---	0.081955	---	---
h=2	5.317065	---	5.505497	---	---
h=3	12.948060	0.0115	13.552730	0.0089	4
h=4	16.374050	0.0373	17.232500	0.0278	8
h=5	18.099770	0.1127	19.121020	0.0856	12

* Test is valid only for lags larger than the VAR lag order.

[!!] WARNING: Significant autocorrelation detected. VAR residuals are not white noise — results may be unreliable.

VAR Residual Serial Correlation (LM) Tests

VAR(2): TFP_unadj ~ EFF | T=58 (1962–2019)

Null: No serial correlation at lag h

Lag	LRE* stat	df	Prob.	Rao F–stat	df	Prob.
h=1	2.885638	4	0.5771	0.724568	(4, 100.0)	0.5772
h=2	6.536641	4	0.1625	1.671486	(4, 100.0)	0.1625
h=3	8.032392	4	0.0904	2.069414	(4, 100.0)	0.0904
h=4	3.810902	4	0.4322	0.961315	(4, 100.0)	0.4322
h=5	0.764440	4	0.9432	0.189936	(4, 100.0)	0.9432

Null: No serial correlation at lags 1 to h

Lag	LRE* stat	df	Prob.	Rao F–stat	df	Prob.
h=1	2.885638	4	0.5771	0.724568	(4, 100.0)	0.5772
h=2	12.415150	8	0.1336	1.603298	(8, 96.0)	0.1339
h=3	18.520010	12	0.1008	1.612854	(12, 92.0)	0.1015
h=4	28.044920	16	0.0312	1.888744	(16, 88.0)	0.0320
h=5	31.176690	20	0.0529	1.672719	(20, 84.0)	0.0548

* Edgeworth expansion corrected likelihood ratio statistic.

VAR(2): TFP_adj ~ EFF | T=58 (1962–2019)

Null: No serial correlation at lag h

Lag	LRE* stat	df	Prob.	Rao F–stat	df	Prob.
h=1	2.578185	4	0.6307	0.646380	(4, 100.0)	0.6307
h=2	5.296216	4	0.2582	1.345925	(4, 100.0)	0.2583
h=3	8.146066	4	0.0864	2.099897	(4, 100.0)	0.0864
h=4	3.704362	4	0.4475	0.933944	(4, 100.0)	0.4475
h=5	1.736923	4	0.7840	0.433650	(4, 100.0)	0.7840

Null: No serial correlation at lags 1 to h

Lag	LRE* stat	df	Prob.	Rao F–stat	df	Prob.
h=1	2.578185	4	0.6307	0.646380	(4, 100.0)	0.6307
h=2	16.128220	8	0.0406	2.123187	(8, 96.0)	0.0407
h=3	21.792290	12	0.0399	1.931237	(12, 92.0)	0.0403
h=4	27.305590	16	0.0382	1.831465	(16, 88.0)	0.0390
h=5	29.110790	20	0.0856	1.543701	(20, 84.0)	0.0881

* Edgeworth expansion corrected likelihood ratio statistic.

[OK] No significant serial correlation detected (LM, all valid lags $p > 0.05$). Green light for cointegration.

VAR Residual Normality Tests

VAR(2): TFP_unadj ~ EFF | T=58 (1962–2019)

Component	Skewness	Chi-sq	df	p	Kurtosis	Chi-sq	df	p	JB stat	df	p
Eq.1	−0.359979	1.252656	1	0.2630	5.283244	12.598570	1	0.0004	13.851230	2	0.0010
Eq.2	−0.310736	0.933385	1	0.3340	3.338524	0.276946	1	0.5987	1.210330	2	0.5460

Test	Statistic	df	Prob
Joint skewness	2.186041	2	0.3352
Joint kurtosis	12.875520	2	0.0016
Joint JB	15.061560	4	0.0046

VAR(2): TFP_adj ~ EFF | T=58 (1962–2019)

Component	Skewness	Chi-sq	df	p	Kurtosis	Chi-sq	df	p	JB stat	df	p
Eq.1	0.371532	1.334350	1	0.2480	4.470216	5.223706	1	0.0223	6.558056	2	0.0377
Eq.2	−0.676337	4.421838	1	0.0355	4.029978	2.563734	1	0.1093	6.985572	2	0.0304

Test	Statistic	df	Prob
Joint skewness	5.756188	2	0.0562
Joint kurtosis	7.787440	2	0.0204
Joint JB	13.543630	4	0.0089

[!!] WARNING: Non-normal residuals detected (joint JB p < 0.05). Cointegration p-values may be affected.

Johansen Cointegration Tests: Hz (no deterministic term)

TFP_unadj ~ EFF | Hz (no deterministic term) | Lags (1st diff.): 1 to 2 | T=53 (1962–2014)

Unrestricted Cointegration Rank Test (Trace)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0) *	0.451741	34.31000	20.16000	0.0002
At most 1 (r=1)	0.060348	3.24680	9.14000	0.7002

* Trace indicates 1 CE at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Unrestricted Cointegration Rank Test (Max–Eigenvalue)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0) *	0.451741	31.25241	14.90000	0.0001
At most 1 (r=1)	0.060348	3.24680	8.18000	0.3906

* Max–Eigenvalue indicates 1 CE at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Normalised Cointegrating Coefficients (std. error in parentheses):

Variable	Coefficient	Std. Error	beta=2.5430 > 0 Valid
TFP_unadj	1.000000		
EFF	−2.543000	(0.26800)	

TFP_adj ~ EFF | Hz (no deterministic term) | Lags (1st diff.): 1 to 2 | T=53 (1962–2014)

Unrestricted Cointegration Rank Test (Trace)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0) *	0.449102	43.89000	20.16000	0.0001
At most 1 (r=1)	0.140514	9.03000	9.14000	0.0525

* Trace indicates 1 CE at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Unrestricted Cointegration Rank Test (Max–Eigenvalue)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0) *	0.449102	31.00272	14.90000	0.0001
At most 1 (r=1)	0.140514	7.87386	8.18000	0.0558

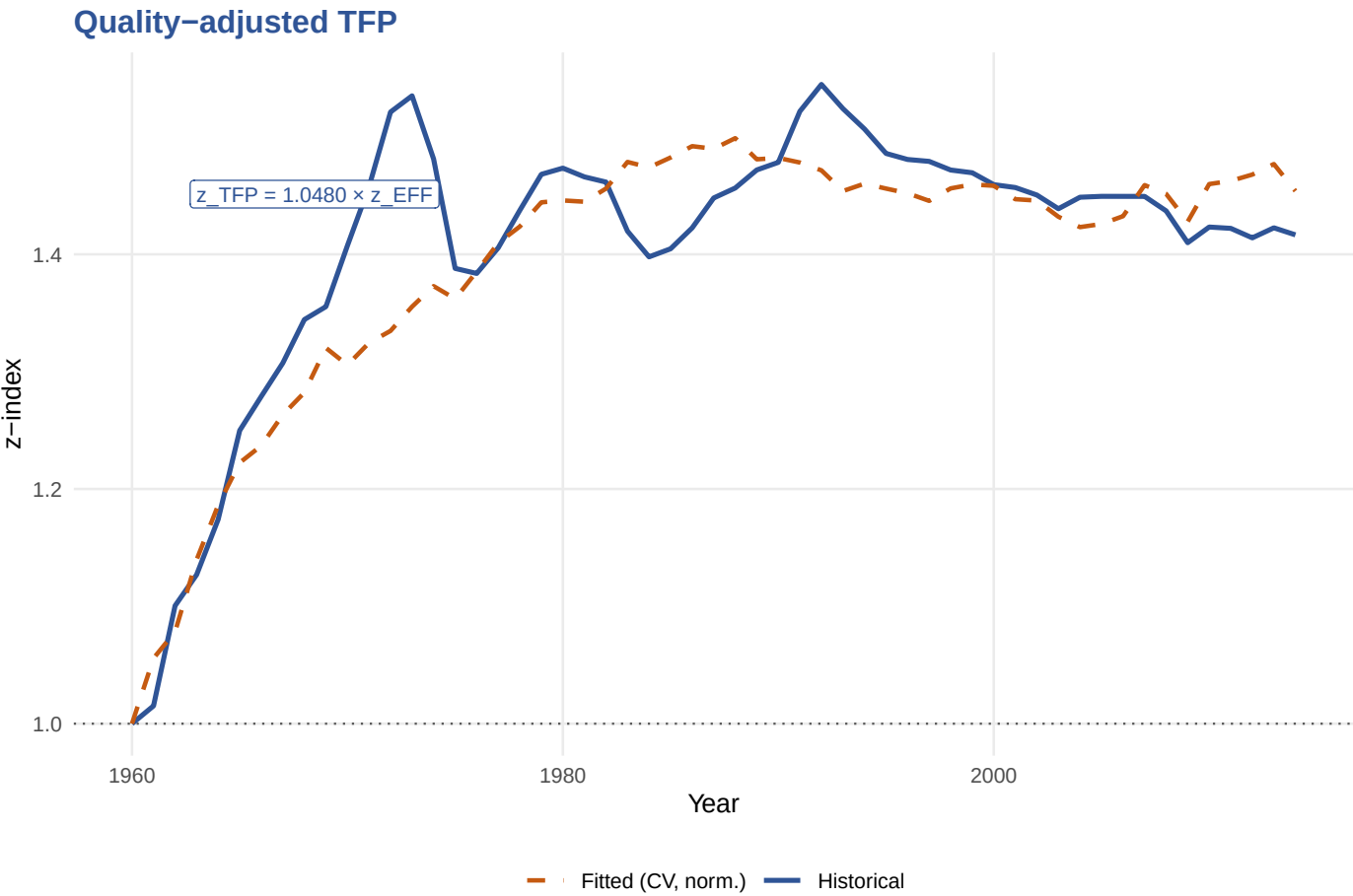
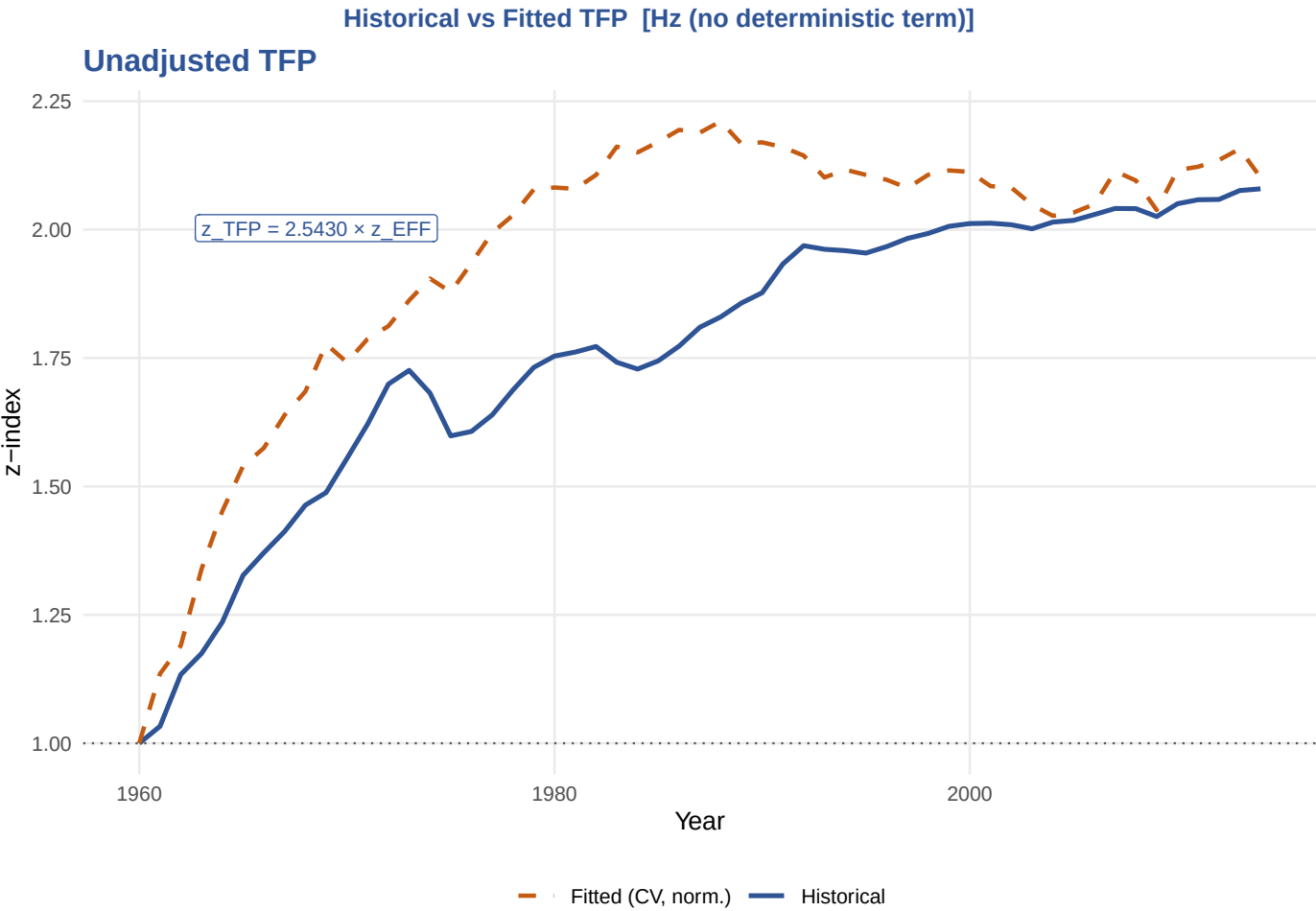
* Max–Eigenvalue indicates 1 CE at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Normalised Cointegrating Coefficients (std. error in parentheses):

Variable	Coefficient	Std. Error	beta=1.0480 > 0 Valid
TFP_adj	1.000000		
EFF	−1.048000	(0.04300)	

Johansen Cointegration Tests: Hc (restricted constant; no trend)

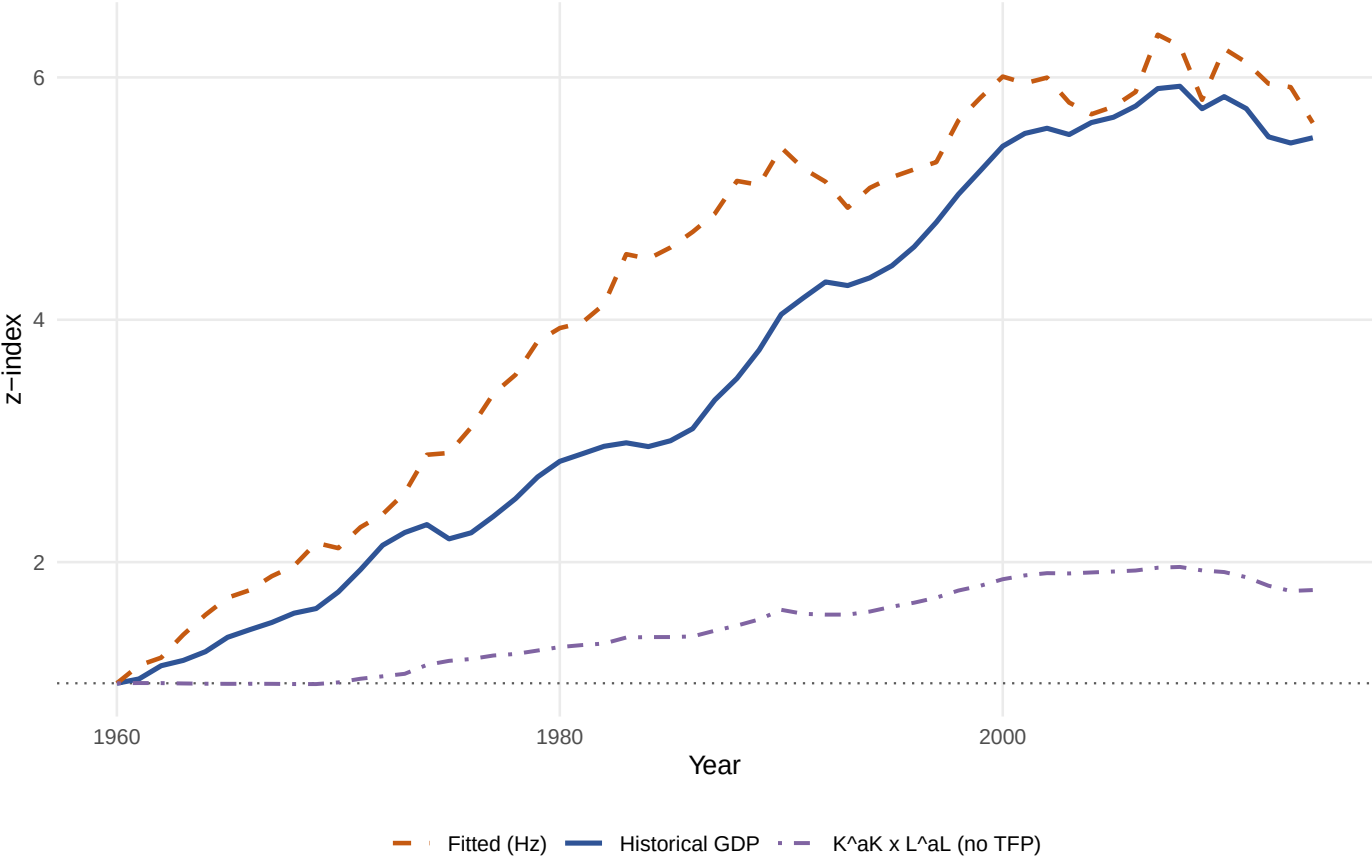


Historical vs Fitted TFP [Hc (restricted constant; no trend)]

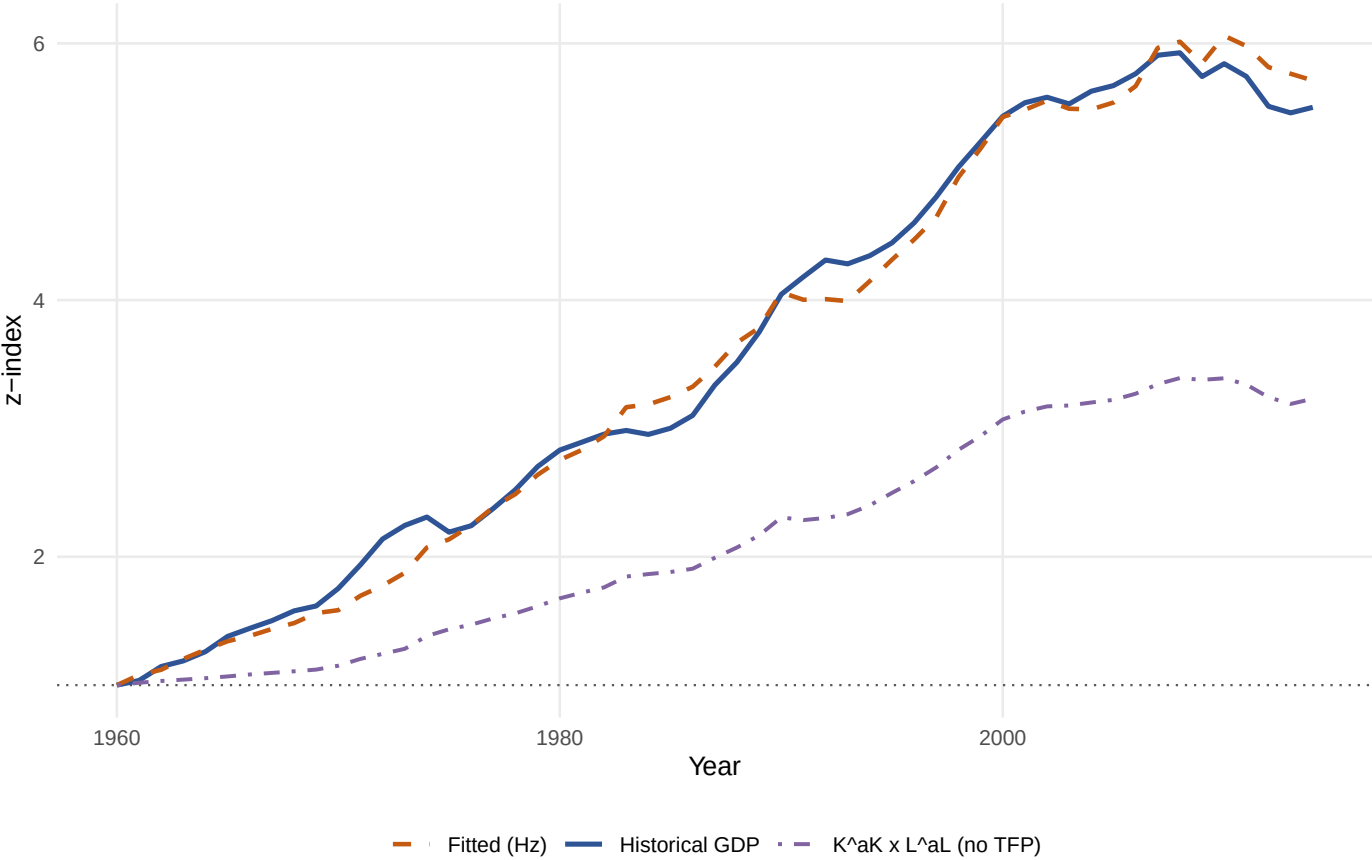
N/A: Unadjusted TFP

N/A: Quality-adjusted TFP

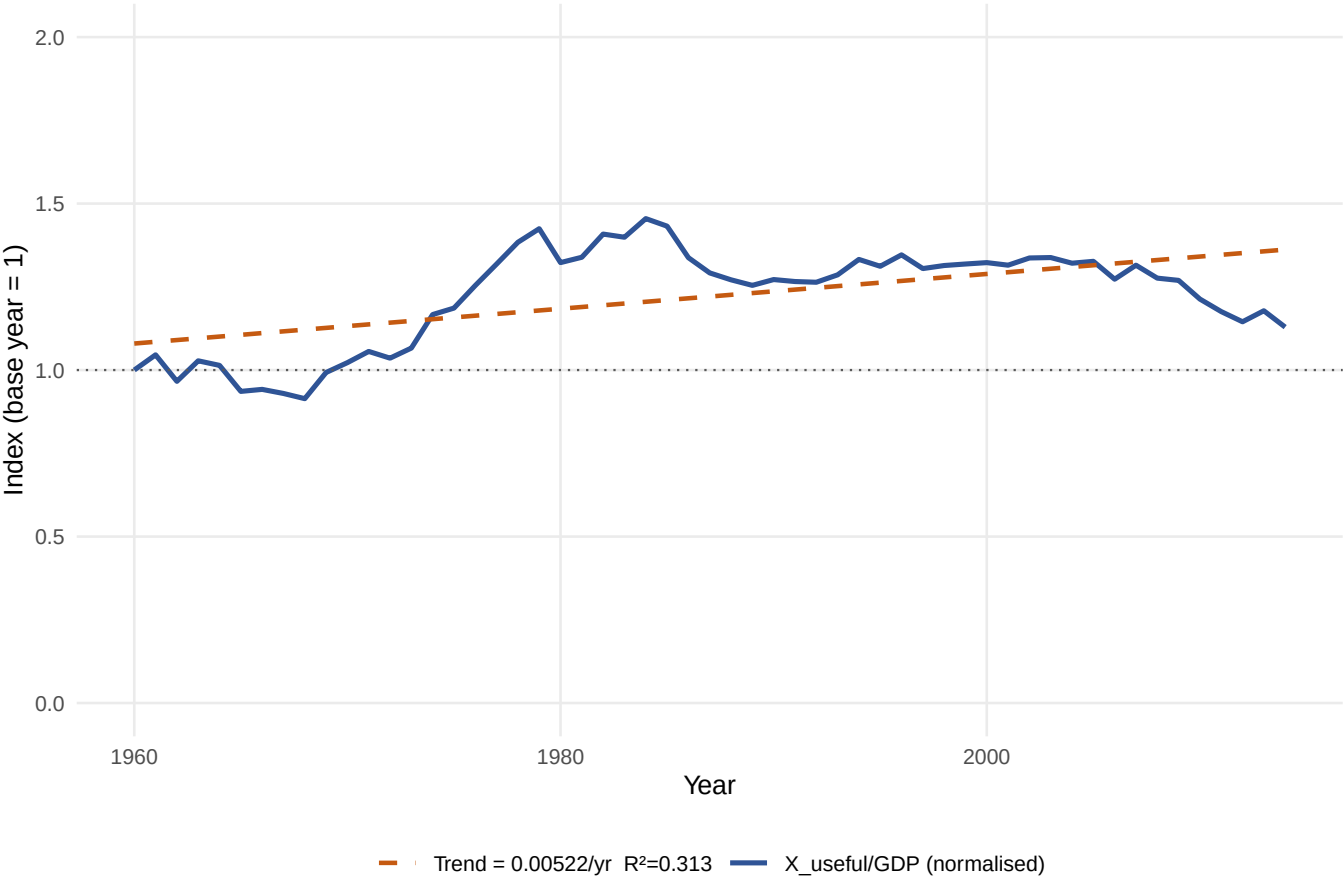
Historical vs Fitted GDP [Hz specification | $K^{\alpha_K} \times L^{\alpha_L}$ = factor accumulation without TFP]
Unadjusted (K_{stock}, L)



Quality-adjusted ($K_{services}, hL$)



Useful Exergy Intensity [X_useful/GDP, normalised to base year = 1]



Statistic	Full (1960–2014, n=55)	Sub (1996–2014, n=19)
Mean intensity	0.9986	1.0426
Std deviation	0.1213	0.0559
CV (%)	12.15	5.36
Min	0.7477	0.9400
Max	1.1901	1.0960
OLS slope (per year)	0.004274	−0.008630
Trend t-stat	9.689	−12.577
Trend p-value	0.0000	0.0000
Trend R2	0.3127	0.7155
Annual growth rate (% p.a.)	0.428	−0.828
ADF statistic (levels)	−2.1200	−2.8500
ADF CV 5%	−3.4925	−3.0820
PP statistic (levels)	−2.0800	−2.7000
PP CV 5%	−3.4862	−3.0820

Individual test results

Statistic	Full (1960–2014, n=55)	Sub (1996–2014, n=19)
No significant trend (p>0.10)	No	No
ADF: I(0) at 5%	No	No
PP: I(0) at 5%	No	No
CV < 5%	No	No
VERDICT: Approx. constant?	No	No

References

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